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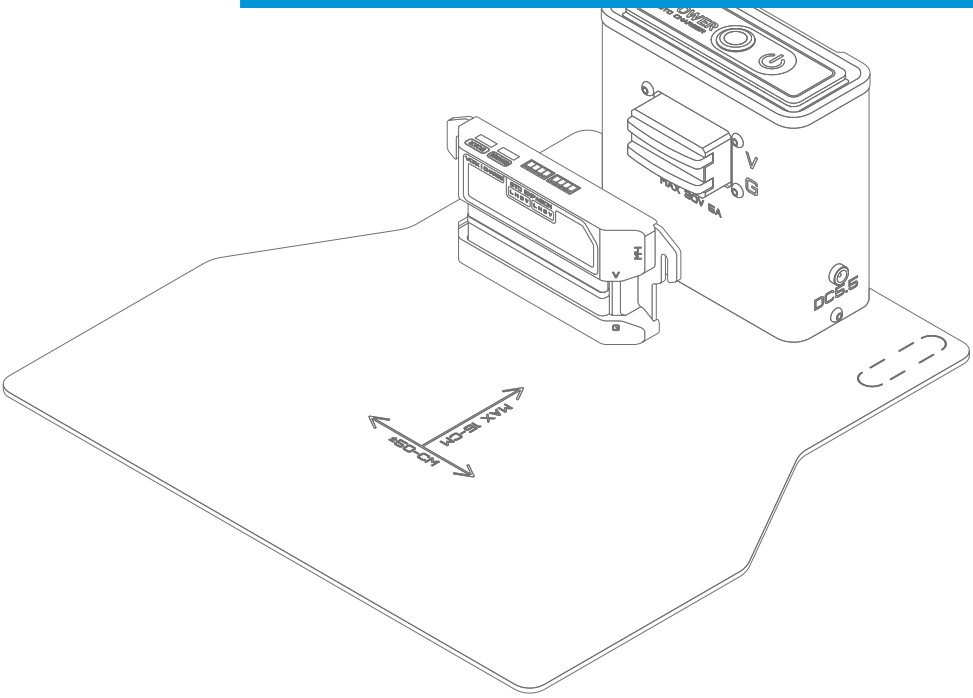
auto charge

Automatic charging station

User Manual

v1.1.1



Important Safety Information

- Please confirm the device's operating voltage, power, installation and other parameters before use. Problems caused by excessive parameters are not covered by the warranty.
- Please evaluate the IP level, temperature and suitability of the equipment in the use environment. Problems caused by harsh environments are not covered by the warranty.
- The device does not provide stable related safety functions such as anti-collision, anti-fall, biological proximity warning, etc. Please conduct a safety assessment on the integrated product.

  assessment, and comply with corresponding regulations and certifications to ensure that the developed products have no major safety hazards.
- Please read the relevant maintenance requirements to prevent irreversible damage caused by incorrect operation, such as: battery over-discharge, low tire pressure

  Low pressure causes hub damage, lack of lubrication causes shaft wear, etc.
- For the first use, please place the device in a safe and open environment without any heavy objects, start it according to the operating instructions, and test all relevant

  If you have any questions about the function, please contact the purchase channel service personnel.
- When the equipment is abnormal or an accident occurs, please use emergency stop or turn off the power immediately to avoid secondary damage, and contact relevant technical personnel.

  Do not dismantle by yourself.

Note: Any damage caused by self-disassembly, modification, abnormal use, natural disasters and other force majeure will not be covered by the warranty.

This product does not bear corresponding responsibility for the safety accidents caused by the equipment after user integration. Please evaluate the risks and reliability tests by yourself. This is hereby declared.

Version update information

V1.0.0: Starting version

V1.0.3: Added charger instructions

V1.0.4: Modify instance accumulation and error

V1.0.5: Modify the data bit error

V1.1.0: Added setting number protocol, modified accessory size

V1.1.1: Modified base plate size and accessory size

Product Introduction

Indoor mobile robots often have complete autonomous driving functions and are often used in unattended working environments. Manual charging can no longer meet the needs of fully automatic operation. In order to support a variety of chassis, we have developed a modular automatic charging station AUTO-CHARGE system to facilitate users to quickly develop fully automatic services.

AUTO-CHARGE It is a modular, quick-disassembled, controllable automatic charging system.

Modularity: The vehicle-mounted end of the system is fully packaged and can support a variety of chassis models, and can upgrade manual charging equipment to automatic

Enter

Support devices

Software Platform

Ubuntu 18
Ubuntu 20

ROS
Melodic
Noetic

ROS2
Galactic
Foxy

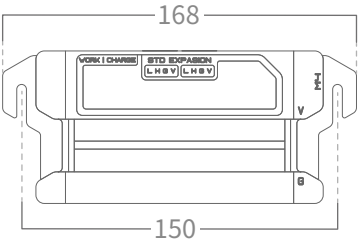
Product Parameters

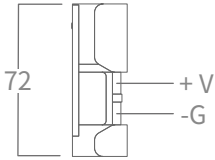
model	AUTO CHARGE Car Terminal
Maximum input voltage	33V
Maximum input current	5A
Power consumption	1.2W
Working Mode	Manual/Automatic
Communication	CAN bus
Safety measures	Contact voltage detection
Operating temperature	- 20~60°C
IP Rating	P22

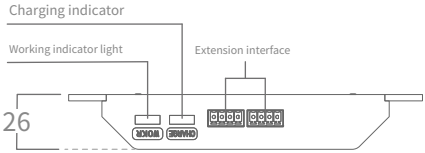
model	AUTO CHARGE Base Station
Transformer input voltage	220V AC
Transformer maximum output voltage	33V DC
Transformer power interface	DC5.5
Contact compressible distance	15MM
Contact left and right deviation distance	±50MM
Safety measures	Contact compression distance detection
Operating temperature	- 20~60°C
IP Rating	P22

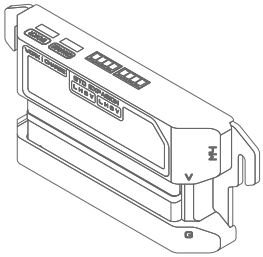
Note: This device uses an external transformer. Please select the corresponding transformer according to the chassis battery requirements.

Vehicle terminal size

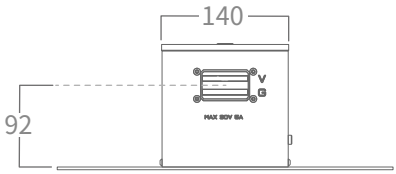


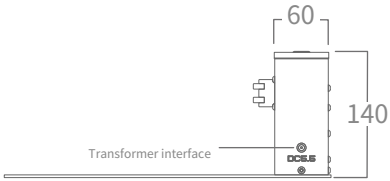


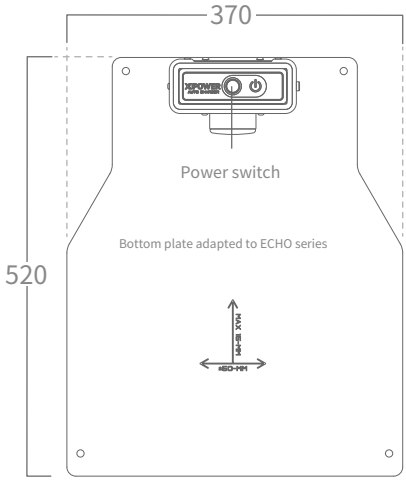


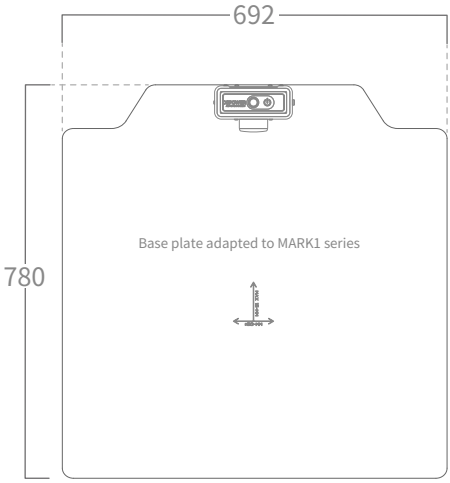


Base station size





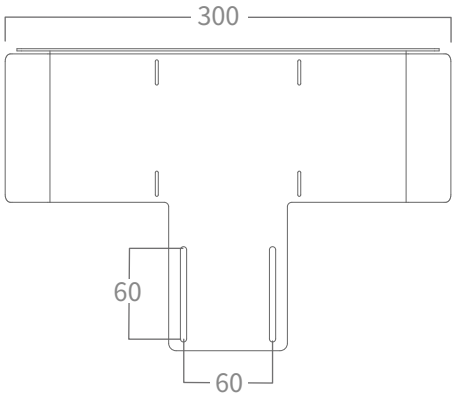


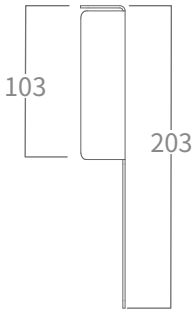


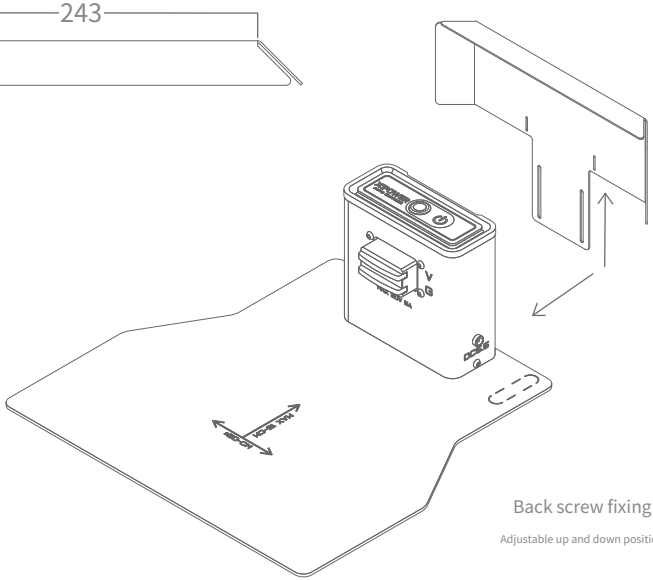
Accessories (optional)

In order to improve the positioning accuracy of the charging station, we provide the charging station with an optional positioning reflector. The height of the reflector can be adjusted up and down to match the installation position of the lidar. Reflective stickers can also be pasted on the reflective board to enhance the effect. If you need special For special sizes, please contact customer service for customized services.

Accessory dimensions and installation







Back screw fixing

Adjustable up and down position

How to use

1. Please connect the output interface of the AC transformer that comes with the robot to the base. The transformer should use the DC5.5 standard. Coaxial interface, maximum output voltage does not exceed 33V.

2. Please press the power switch on the top of the base. The blue light will turn on to indicate that the power is on. The contacts will not be short-circuited when they touch objects. The contacts will only be energized when they retract more than 5mm.

3. Please install the vehicle-mounted terminal at the front or rear end of the chassis, adjust the height to match the contactor, and connect the extension line to the chassis.

Power supply connectedIf CAN communication is required, please connect the H and L lines of the expansion interface.

4. After powering on the vehicle terminal, wait for the buzzer to sound three times, which means the system is working properly. If the buzzer sounds continuously, Please check if there is any error, please see the LED example and the corresponding communication protocol.

Precautions

1. Please place the charging base as close to the wall as possible to prevent the chassis from being displaced due to collision.

2. When not in contact, neither the base nor the vehicle terminal will be charged, but please do not push the base contact points manually.

3. If the laser positioning is inaccurate, please use the included positioning plate.

Extension interface line sequence			
V	G	H	L
Positive power supply	Negative pole of power supply	CAN bus H	CAN bus L

Vehicle terminal working status indication		
Working status	Charging LED Status	Buzzer status
Not charging, not in contact	No Light	none
Not charging, contacted	Green light flashing	none
Charging	Red and green alternating	none
Full	Green light is always on	none
mistake	Red light is on	Interval beep

Working mode status table			
Working Mode	Contact status	Manual switch	Working status
automatic	No contact	invalid	Not charging
	touch	invalid	Charge
Manual	No contact	invalid	Not charging
	touch	connect	Charge
		Disconnect (default)	Not charging

Vehicle terminal indicator light status

The embedded vehicle terminal is integrated into the chassis and can only be queried through the CAN bus. The following content only applies to the external vehicle terminal device

Working indicator light:

During the self-test or upgrade, the buzzer will sound three times after the end of the test.

Normal working status

Charging indicator:

Not in charging state, not in contact with charging pile

Not charging but already in contact with the charging station

Charging

Charging complete

An error occurred

Charging station indicator status

The charging pile indicator light is located on the ring of the power switch button

Powered on

The vehicle terminal has been touched, but charging has not started or has been completed.

Charging

Charging precautions

Lithium batteries require different charging voltages depending on the number of battery strings. Please choose the correct transformer for charging.

Use the device's own transformer. If you have any questions, please contact customer service.

Working indicator light:

Not charging or charging completed

Charging

Precautions

1. Please make sure that the charging voltage is correct.The device charging voltage will be marked on the device nameplate.

2. Please make sure to charge the device when there are people around. Please cut off the power supply when no one is around.

3. Please start charging after the device is docked. Do not use the device while charging.

4. The charger and equipment should be kept away from the bedroom, with no flammable materials around them and kept dry.

5. The charger uses 220V AC input power, please ensure that the input voltage is stable.

6. Single continuous charging time should not exceed 8 hours. If the charger overheats, stop charging immediately.

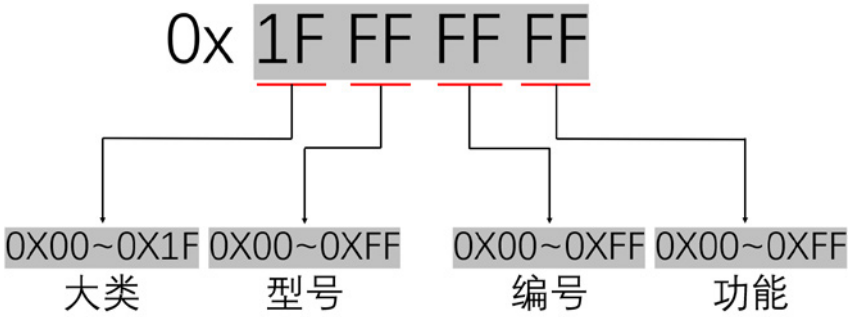
Using the wrong charger will cause irreversible damage to the battery. Please confirm the charging voltage on the device nameplate before use.

This product does not assume any responsibility for the consequences caused by users' incorrect use of the charger or unsupervised charging for a long time. Users are requested to evaluate the risks and reliability tests on their own. This is hereby declared.

CAN communication protocol

The CAN communication in the product adopts CAN2.0B (extended frame)the communication baud rate is 500K.

CAN ID encoding specification:



CAN data bit: Byte[0]~Byte[7]:

Byte[0]	Byte[1]	Byte[2]	Byte[3]	Byte[4]	Byte[5]	Byte[6]	Byte[7]

The CAN information used by CAN is the CANID and CAN_DATA mentioned above, totaling 12 bytes of data. When using, the needle Perform corresponding CAN_DATA data transmission for different CANIDs.

The category and model in CANID are fixed parameters of the device. The value will not change. The default number is 1. When multiple devices of the same type are used at the same time, they can be modified through general instructions.

CAN data frame					
Data bits	ID[3]	ID[2]	ID[1]	ID[0]	DATA[0~7]
Data Types	unsigned int8	unsigned int8	unsigned int8	unsigned int8	8 x unsigned int8
Data Description	Major categories	model	serial number	Function	8 bytes of CAN data

This equipment is fixed in the following categories 06

Fixed model 02

Default Number 01

The fixed format of CANID of this device is 06 02 01 XX

Instruction Set Query Description

The CAN protocol is divided into general device commands and device-specific commands. General command operations are applicable to all module accessories of the system, and are commonly used for device startup, system upgrade, version query, system reset, etc.; device-specific commands are exclusive to different device modules, and are mainly used for restarting and clearing errors for charging stations.

协议目录

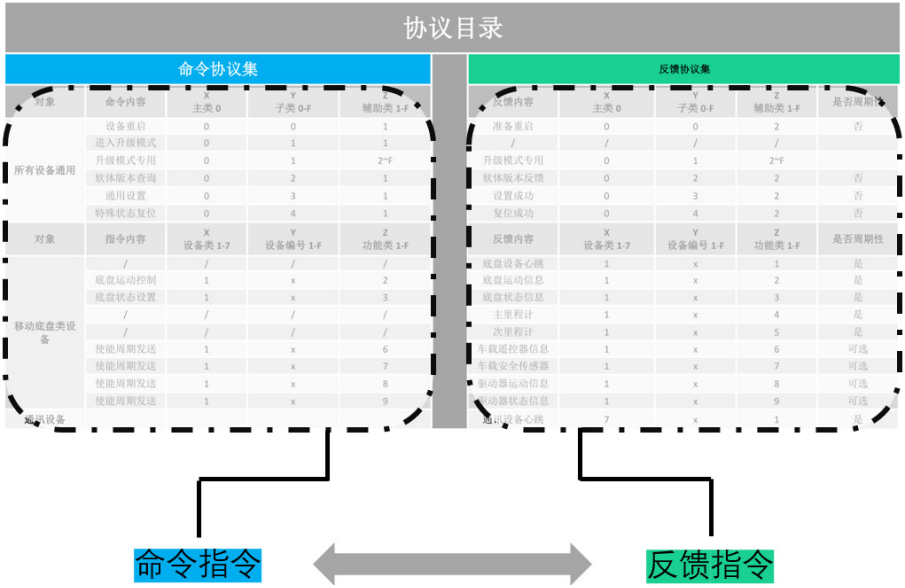
命令协议集

对象	命令内容	X 主类 0	Y 子类 0-F	Z 辅助类 1-F
所有设备通用	设备重启	0	0	1
	进入升级模式	0	1	1
	升级模式专用	0	1	2~F
	软体版本查询	0	2	1
	通用设置	0	3	1
	特殊状态复位	0	4	1
对象	指令内容	X 设备类 1-7	Y 设备编号 1-F	Z 功能类 1-F
移动底盘类设备	/	/	/	/
	底盘运动控制	1	x	2
	底盘状态设置	1	x	3
	/	/	/	/
	/	/	/	/
	使能周期发送	1	x	6
	使能周期发送	1	x	7
	使能周期发送	1	x	8
	使能周期发送	1	x	9
通讯设备				

反馈协议集

反馈内容	X 主类 0	Y 子类 0-F	Z 辅助类 1-F	是否周期性
准备重启	0	0	2	否
/	/	/	/	
升级模式专用	0	1	2~F	
软体版本反馈	0	2	2	否
设置成功	0	3	2	否
复位成功	0	4	2	否
反馈内容	X 设备类 1-7	Y 设备编号 1-F	Z 功能类 1-F	是否周期性
底盘设备心跳	1	x	1	是
底盘运动信息	1	x	2	是
底盘状态信息	1	x	3	是
主里程计	1	x	4	是
次里程计	1	x	5	是
车载遥控器信息	1	x	6	可选
车载安全传感器	1	x	7	可选
驱动器运动信息	1	x	8	可选
驱动器状态信息	1	x	9	可选
通讯设备心跳	7	x	1	是

CAN closed-loop control system has a response mechanism for most commands. That is, the command frame and the feedback frame correspond Based on the feedback frame Monitor data stream accuracy in real time.



协议目录

命令协议集

对象	命令内容	X 主类 0	Y 子类 0-F	Z 辅助类 1-F
所有设备通用	设备重启	0	0	1
	进入升级模式	0	1	1
	升级模式专用	0	1	2~F
	软体版本查询	0	2	1
	通用设置	0	3	1
	特殊状态复位	0	4	1
对象	指令内容	X 设备类 1-7	Y 设备编号 1-F	Z 功能类 1-F
移动底盘类设备	/	/	/	/
	底盘运动控制	1	x	2
	底盘状态设置	1	x	3
	/	/	/	/
	/	/	/	/
	使能周期发送	1	x	6
	使能周期发送	1	x	7
	使能周期发送	1	x	8
	使能周期发送	1	x	9
通讯设备				

反馈协议集

反馈内容	X 主类 0	Y 子类 0-F	Z 辅助类 1-F	是否周期性
准备重启	0	0	2	否
/	/	/	/	
升级模式专用	0	1	2~F	
软体版本反馈	0	2	2	否
设置成功	0	3	2	否
复位成功	0	4	2	否
反馈内容	X 设备类 1-7	Y 设备编号 1-F	Z 功能类 1-F	是否周期性
底盘设备心跳	1	x	1	是
底盘运动信息	1	x	2	是
底盘状态信息	1	x	3	是
主里程计	1	x	4	是
次里程计	1	x	5	是
车载遥控器信息	1	x	6	可选
车载安全传感器	1	x	7	可选
驱动器运动信息	1	x	8	可选
驱动器状态信息	1	x	9	可选
通讯设备心跳	7	x	1	是

Protocol Directory

Command Protocol Set						Feedback Protocol Set					
Object	Command content	Major categories 00~1F	model 00~FF	serial number 00~FF	Function 00~FF	Command content	Major categories 00~1F	model 00~FF	serial number 00~FF	Function 00~FF	Periodicity
All equipment General	Device restart	06	02	01	01	Preparing for Restart	06	02	01	A1	no
	General settings	06	02	01	03	General settings successful	06	02	01	A3	no
	Special state reset	06	02	01	04	Special reset successful	06	02	01	A4	no
	Numbering settings	06	02	01	06	Set success feedback	06	02	01	A6	no
	/	/	/	/	/	Heartbeat Packet	06	02	01	B0	yes
Object	Command content	bigind 00~1F	model 00~FF	serial number 00~FF	Function 00~FF	Command content	Major categories 00~1F	model 00~FF	serial number 00~FF	Function 00~FF	Periodicity
Automatic charging station	Charging station setup	06	02	01	13	Charging station status information	06	02	01	B3	yes
	/	/	/	/	/	Charging station data feedback	06	02	01	B4	yes

General instructions for equipment

Device restart/prepare to restart

Device restart				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
external	This equipment	0x06 02 01 01	8	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Major categories	unsigned int8	Always 0x06	
byte[1]	model	unsigned int8	Always 0x02	
byte[2]	serial number	unsigned int8	Default is 0x01	
byte[3~7]	/	/	/	

Note: When there are multiple machines in the system, the device number can be changed and set. The default device number is 0x01. The length of the sent command data is 8, and the invalid bits are filled with 0 (the same below).

Preparing for Restart				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 A1	0	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0~7]	/	/	/	

Note: The device restart command restarts the system. When the device receives the command, it responds with a ready-to-restart feedback. The system restarts. After two seconds, the buzzer sounds three times, and the restart is complete.

Example: Send 06 02 01 00 00 00 00 00 to system ID: 0x06 02 01 01, and the system restarts.

Send Example					
Transfer Target	CAN ID	Frame format	Frame Type	Data length	Data bit (L->H)
equipment	0x06 02 01 01	Data Frame	Extended Frame	8	06 02 01 00 00 00 00 00
Host computer	0x06 02 01 A1	Data Frame	Extended Frame	0	/

General Settings/General Settings Success

General settings				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
external	This equipment	0x06 02 01 03	8	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Major categories	unsigned int8	Always 0x06	
byte[1]	model	unsigned int8	Always 0x02	
byte[2]	serial number	unsigned int8	Default is 0x01	
byte[3]	Enable	unsigned int8	0x00: Turn off the device / 0x01: Turn on the device	
byte[4~7]	/	/	/	

General settings successful				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 A3	0	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0~7]	/	/	/	

Note: General setting instructions are used to enable the device.

The charging station is enabled by default when it is powered on.

For example: To enable the device number 0x01, send the command 06 02 01 01 00 00 00 00 to the system ID: 0x06 02 01 03; the device is enabled.

Send Example					
Transfer Target	CAN ID	Frame format	Frame Type	Data length	Data bit (L->H)
equipment	0x06 02 01 03	Data Frame	Extended Frame	8	06 02 01 01 00 00 00 00
Host computer	0x06 02 01 A3	Data Frame	Extended Frame	0	/

Special state reset / Special reset successful

Special state reset				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
external	This equipment	0x06 02 01 04	8	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	State Clearance	unsigned int8	0xCC: Clear/No operation for other values	
byte[1~7]	/	/	/	

Special reset successful				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 A4	0	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0~7]	/	/	/	

Note: Special status reset is used to clear errors and reset the system when it is abnormal.

Example: Send command CC 00 00 00 00 00 00 00 to system ID: 0x06 02 01 04.

Send Example					
Transfer Target	CAN ID	Frame format	Frame Type	Data length	Data bit (L->H)
equipment	0x06 02 01 04	Data Frame	Extended Frame	8	CC 00 00 00 00 00 00 00
Host computer	0x06 02 01 A4	Data Frame	Extended Frame	0	/

Numbering settings

General settings				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
external	This equipment	0x06 02 01 06	8	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Major categories	unsigned int8	Always 0x06	
byte[1]	model	unsigned int8	Always 0x02	
byte[2]	serial number	unsigned int8	Default is 0x01	
byte[3]	New Number	unsigned int8	If the number is not changed, it will be the same as the original number.	
byte[4~7]	/	/		
			/	

General settings successful				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 A6	0	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0~7]	/	/	/	

Note: The number setting command is used to set a new number for the device. When there is only one device in the system, the number is always 0x01 by default. When there are multiple devices in the system, the device number can be customized.

For example: To change the device number to 0x02, send the command 06 02 01 02 00 00 00 00 to the system ID: 0x06 02 01 06; the device number is switched from 0x01 to 0x02.

Send Example					
Transfer Target	CAN ID	Frame format	Frame Type	Data length	Data bit (L->H)
equipment	0x06 02 01 06	Data Frame	Extended Frame	8	06 02 01 02 00 00 00 00
Host computer	0x06 02 01 A6	Data Frame	Extended Frame	0	/

Device heartbeat packet

Heartbeat Packet				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 B0	1	500
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Enable status	unsigned int8	0x00: Disabled / 0x01: Enabled	
byte[1~7]	/	/	/	

Note: The heartbeat packet is mainly used to confirm the online status of the device. It must maintain a sending cycle of 500ms in any state. It is also a key frame to confirm whether the device is enabled.

Device-specific instructions

Charging station settings/charging station status information

Charging station setup				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
external	This equipment	0x06 02 01 13	8	/
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Working Mode	unsigned int8	0x00: Automatic (default) / 0x01: Manual	
byte[1]	Manual switch setting	unsigned int8	0x00: Disconnect (default) / 0x01: Connect	
byte[2]	Buzzer Enable	unsigned int8	0: Disable / 1: Enable (default)	
byte[3]	Charging restart voltage difference①	unsigned int8	Unit: 0.1V (default 1.0V)	
byte[4]	Charging cut-off current②	unsigned int8	Unit: 0.1A (default 0.2A)	
byte[5~7]	/	/	/	

Charging station information				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 B3	8	500/contact trigger③
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Working Mode	unsigned int8	0x00: Automatic / 0x01: Manual	
byte[1]	Contact status	unsigned int8	0x00: Not contacted / 0x01: Contacted	
byte[2]	Working status	unsigned int8	0: Disconnected / 1: Charging / 2: Fully charged (low current) / 3: Error	
byte[3]	Error Status	unsigned int8	View the error status table	
byte[4]	Buzzer enable status	unsigned int8	0: Disable / 1: Enable (default)	
byte[5]	Charging restart voltage difference	unsigned int8	Unit: 0.1V (default 1.0V)	
byte[6]	Charging cut-off current	unsigned int8	Unit: 0.1A (default 0.2A)	
byte[7]	/	/	/	

Note: 1. The charging restart voltage difference refers to the value at which charging will be restarted after the voltage drops to the value after charging is completed.

2. Charging cut-off current means that when the charging current is less than this value, the power supply will be disconnected to complete charging. Setting 0 means permanent disconnection

3. Contact trigger means that when the contacts touch and detect voltage, a command will be sent immediately, allowing users to brake in time

Error Status Table		
byte	Status value	Status Description
byte[3]	0x00	No Error
	0x01	Power supply overvoltage
	0x02	Charging overcurrent
	0x03	Short circuit protection

Charging station data feedback

Charging station information				
Sender	**Receiver**	**CAN ID**	**Data length**	**Period (ms)**
This equipment	external	0x06 02 01 B4	8	500
Data bits	**Function**	**Data Types**	**illustrate**	
byte[0]	Charging voltage low 8 bits	unsigned int16	Unit: 0.01V	
byte[1]	Charging voltage high 8 bits			
byte[2]	Charging current lower 8 bits	unsigned int16	Unit: 0.1A	
byte[3]	Charging current high 8 bits			
byte[4~7]	/	/	/	

Serial communication protocol

If you need to use the computer's USB serial port for communication, you need to use a CAN-COM HUB as a conversion device. Please contact customer service for more technical support.

Serial port parameters	
Baud rate	460800
Check digit	No verification
Data bit length	8
Stop bits	1

Serial data frame					
Data Definition	HEAD	LEN	DATA[0-11]	FRAME ID	SUM
Data Types	uint8	uint8	12 x uint8	uint8	uint8
Data Description	0x55	0x10	12-bit CAN message data Encapsulation reference CAN data frame	Frame ID can be fixed or cumulative Sequence stamps for consecutive data	The cumulative sum of the first 15 contents Then take the lowest uint8 feedback

Note: To simplify system design and improve system reuse rate, keep the serial port byte2 -byte13 consistent with the ten-bit data used in the CAN protocol, see CAN data frame (the command function content is completely consistent with the CAN protocol). The start bit, frame length, and frame ID are the default values, and the checksum is the value of the sum of the first 15 bytes and the last byte of data.

Send Example						
Start bit	frame length	CAN ID	CAN DATA	Frame ID	Checksum	Function
55	10	06 02 01 01	06 02 01 00 00 00 00 00	01	79	Device restart
55	10	06 02 01 03	06 02 01 01 00 00 00 00	01	7C	General settings enable
55	10	06 02 01 04	CC 00 00 00 00 00 00 00	01	3F	Clear Status
55	10	06 02 01 13	01 01 00 00 00 00 00 00	01	84	Setting up a manual connection

XVIEW Test System

XVIEW is a test system running on the host computer. It supports online testing and adjustment of various chassis, sensors, expansion devices, etc., and integrates CAN analyzer and serial port assistant. It is a very practical test tool.

Note: XVIEW requires a CAN-COM HUB as a conversion device. Please contact customer service for more technical support.

XROS Support

XROS is an open source development package based on ROS, which helps users quickly use our various devices and products on the ROS platform, reducing the amount of large code development. This code package supports the full set of XSTD protocol standards. Please contact customer service to obtain the development package and instructions for use. Before use, you need to install the Ubuntu operating system and the ROS robot operating system. This ros package is suitable for The ROS and ROS2 versions are as follows. For usage, please refer to the XROS manual.

Note: XROS requires a CAN-COM HUB as a conversion device. Please contact customer service for more technical support.

HEXMAN ROBOTICS

Qiyuan Robotics (Dongguan) Co., Ltd.

WWW.HEXMAN.CN

TEL: +86-0769-22678112 078112

MOBIL: +86-18652867127